

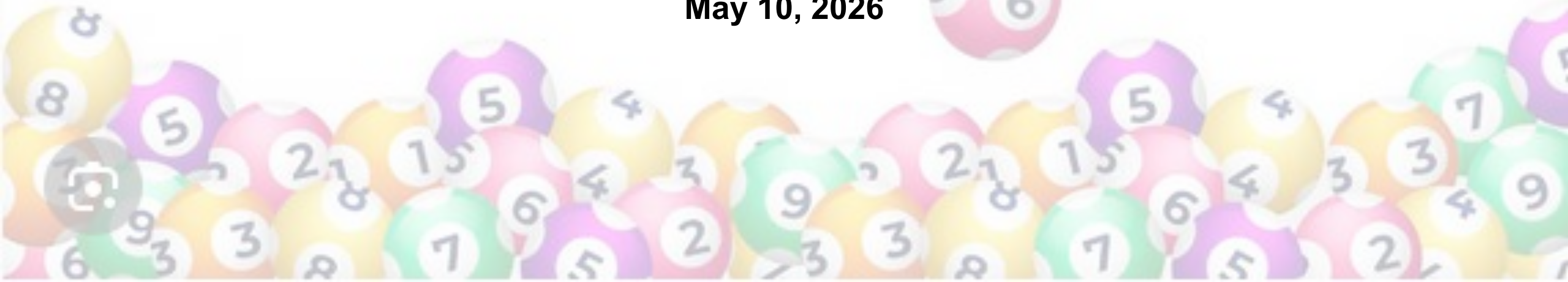


Lottery Number Analysis: Patterns or Randomness?

Analysis of Mega Millions, Powerball, and Pick 10

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Audience and Purpose

Audience

- General Public
- Lottery Participants

Purpose

- Evaluate whether historical lottery numbers reveal meaningful patterns

Callout

- Focus on understanding – not prediction

Data Overview

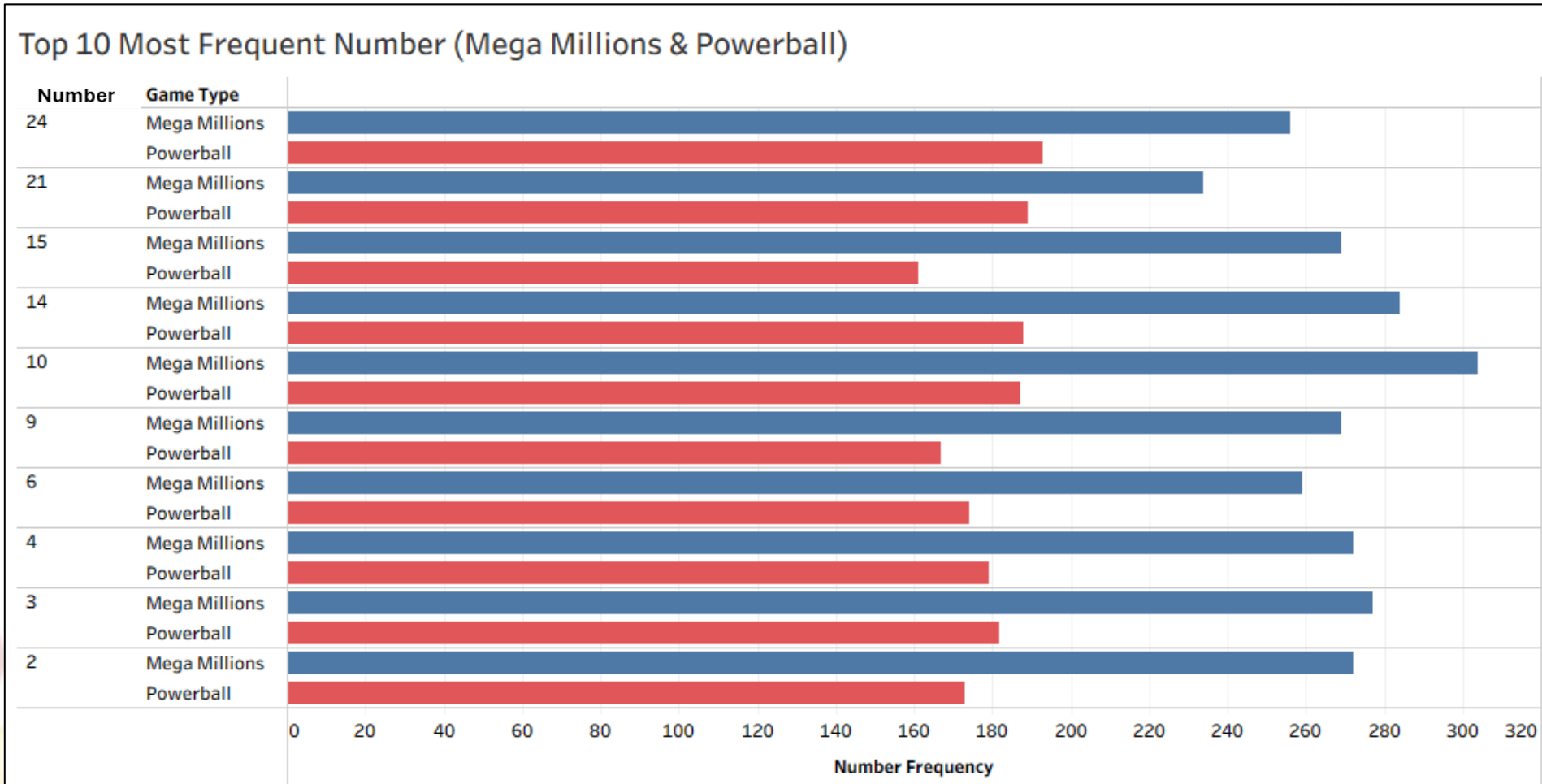
Data Source:

- data.ny.gov

Datasets and Years Covered:

- Mega Millions: 2002 – 2024
- Powerball: 2010 – 2024
- Pick 10: 1987 – 2024

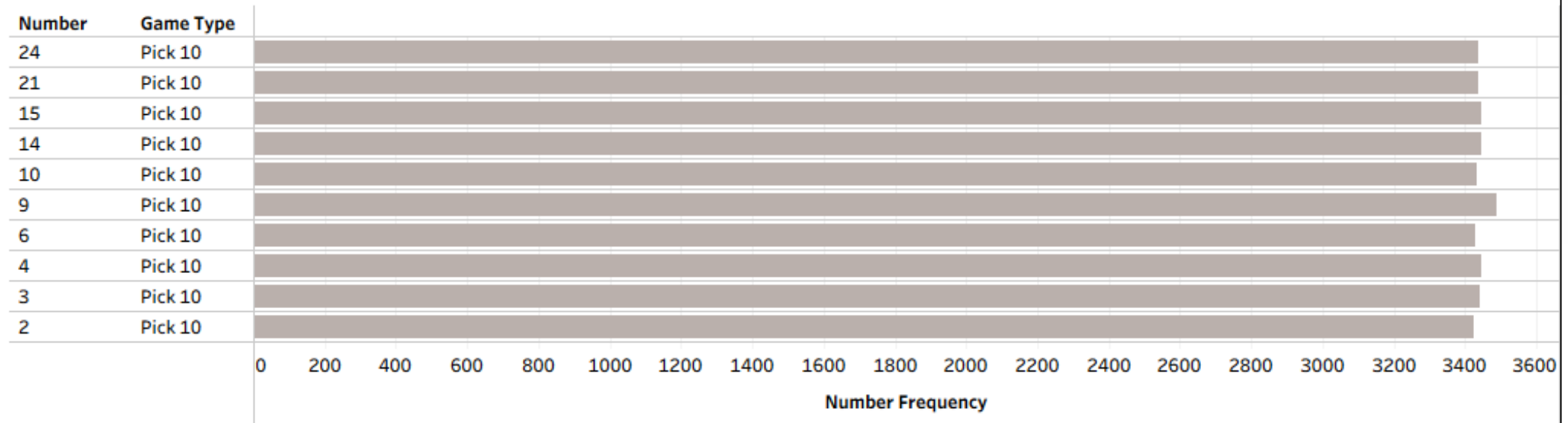
Ten Most Frequent Numbers (Mega Millions & Powerball)



Frequencies are relatively balanced; no number dominates consistently

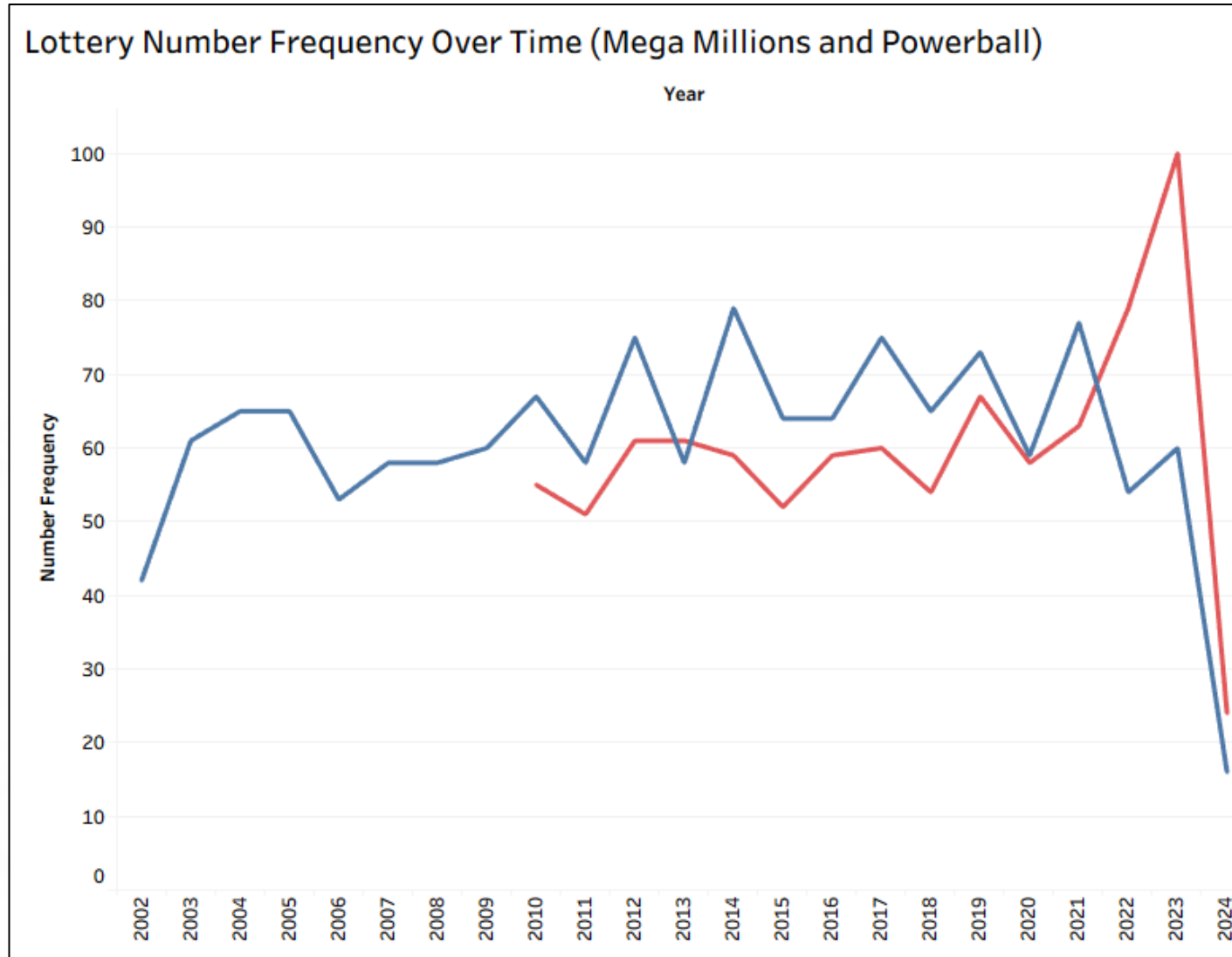
Ten Most Frequent Numbers (Pick 10)

Top 10 Numbers - Pick 10



Higher counts due to 20 numbers per draw – not directly comparable

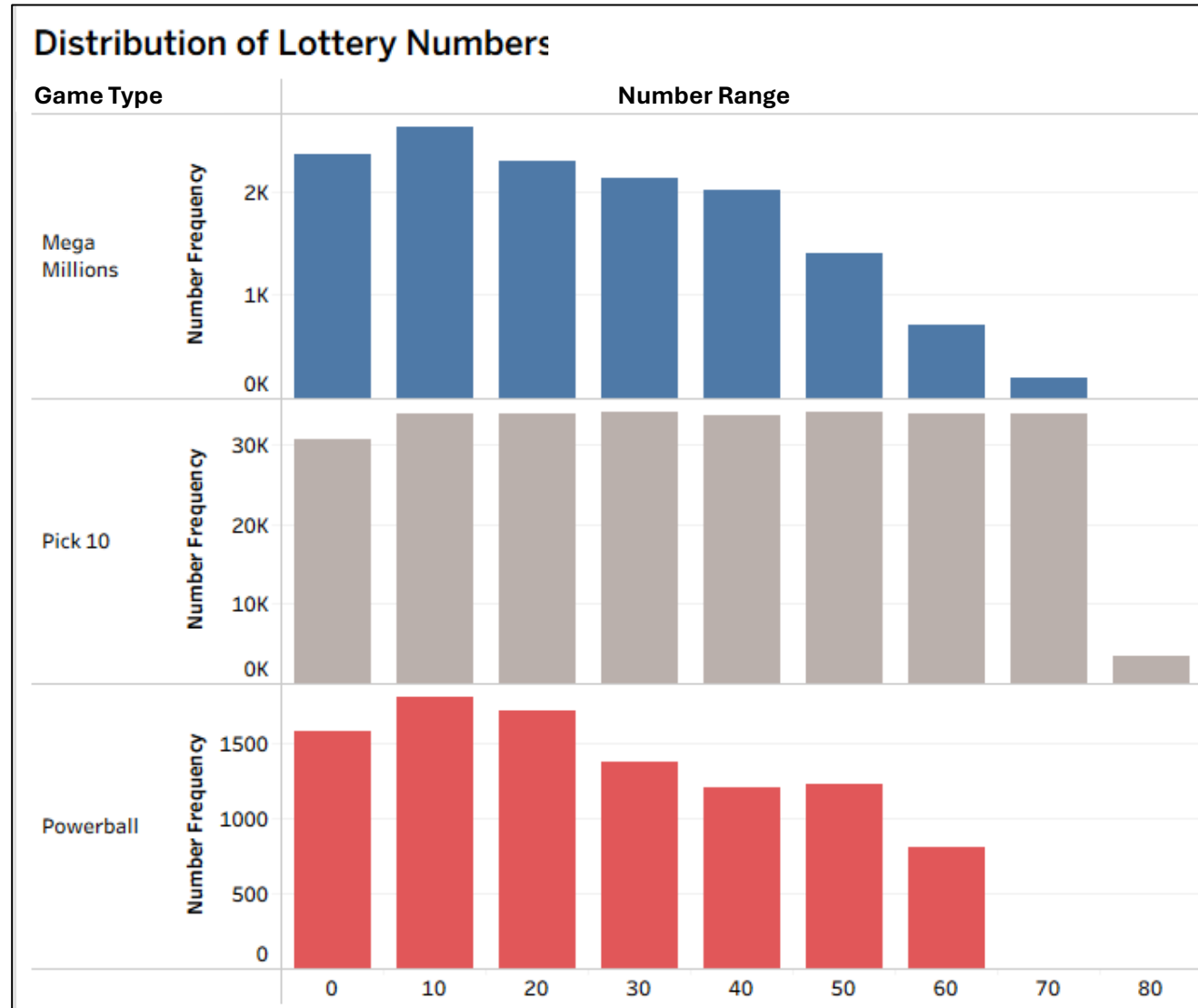
Trends Over Time



No consistent trends; fluctuations appear random

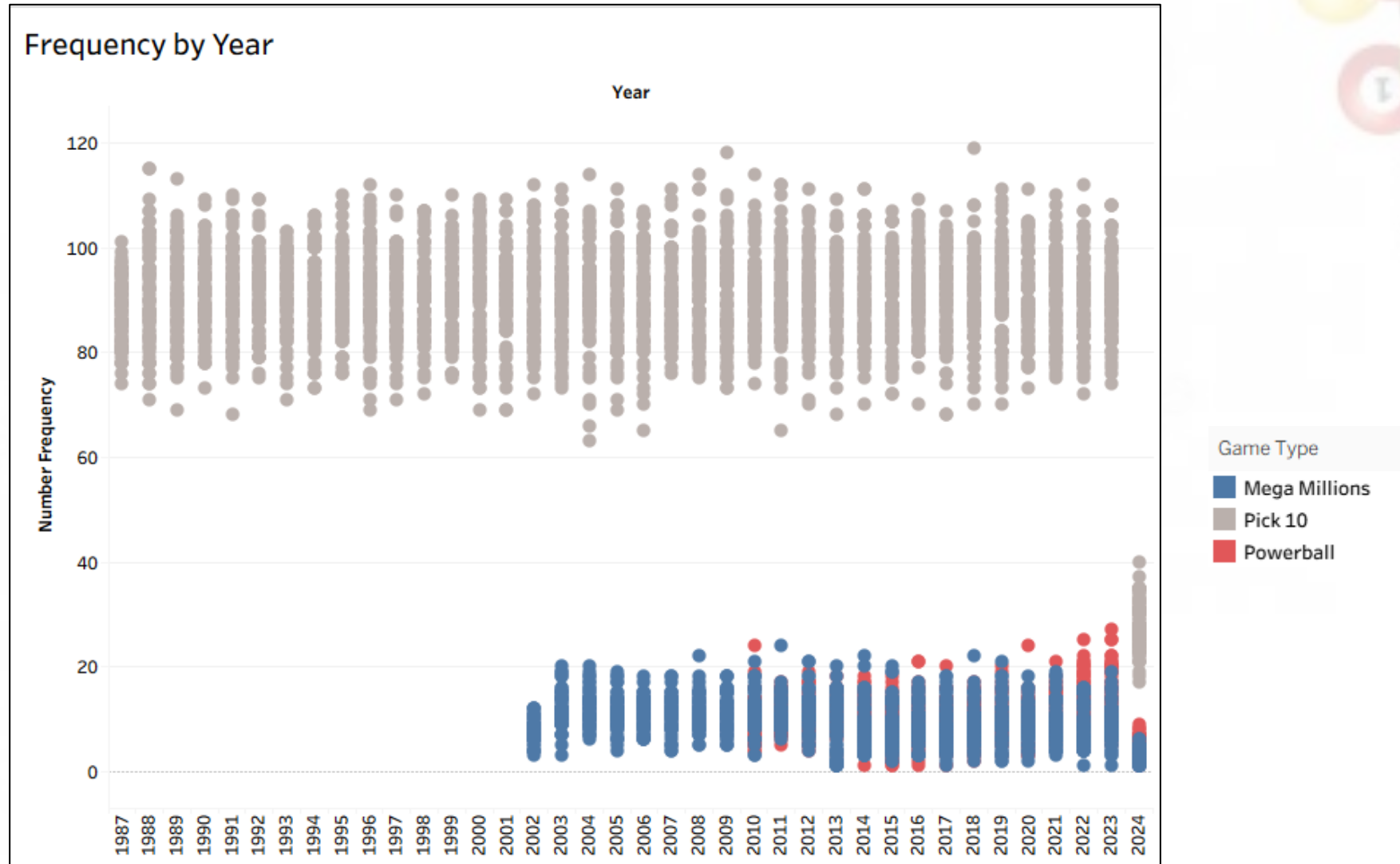
Distribution of Lottery Numbers by Game

Pick 10: 20 numbers
per draw -> higher
counts



Numbers are evenly
distributed – no
clustering

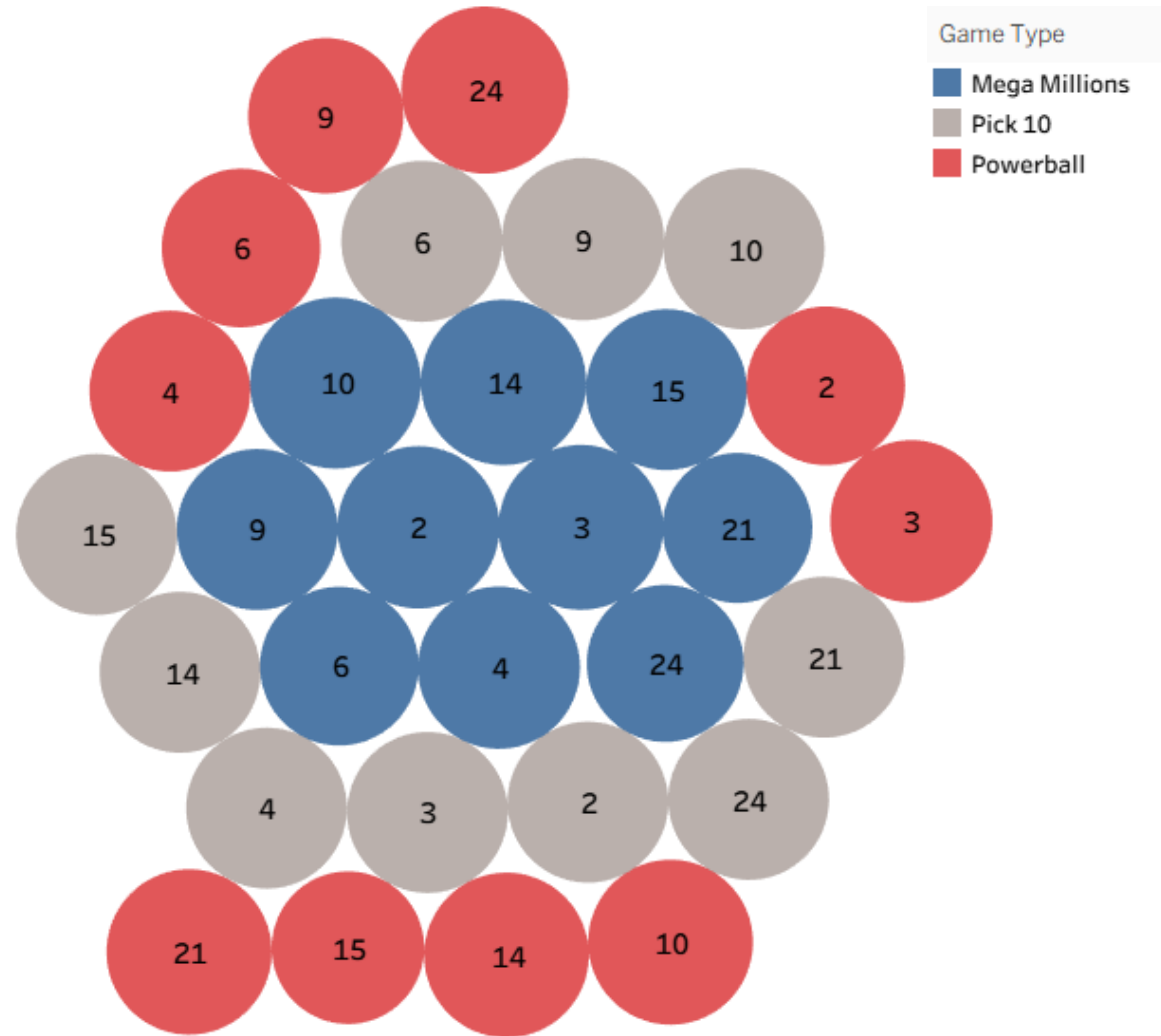
No Clear Patterns Over Time



Points are widely dispersed — no clear pattern

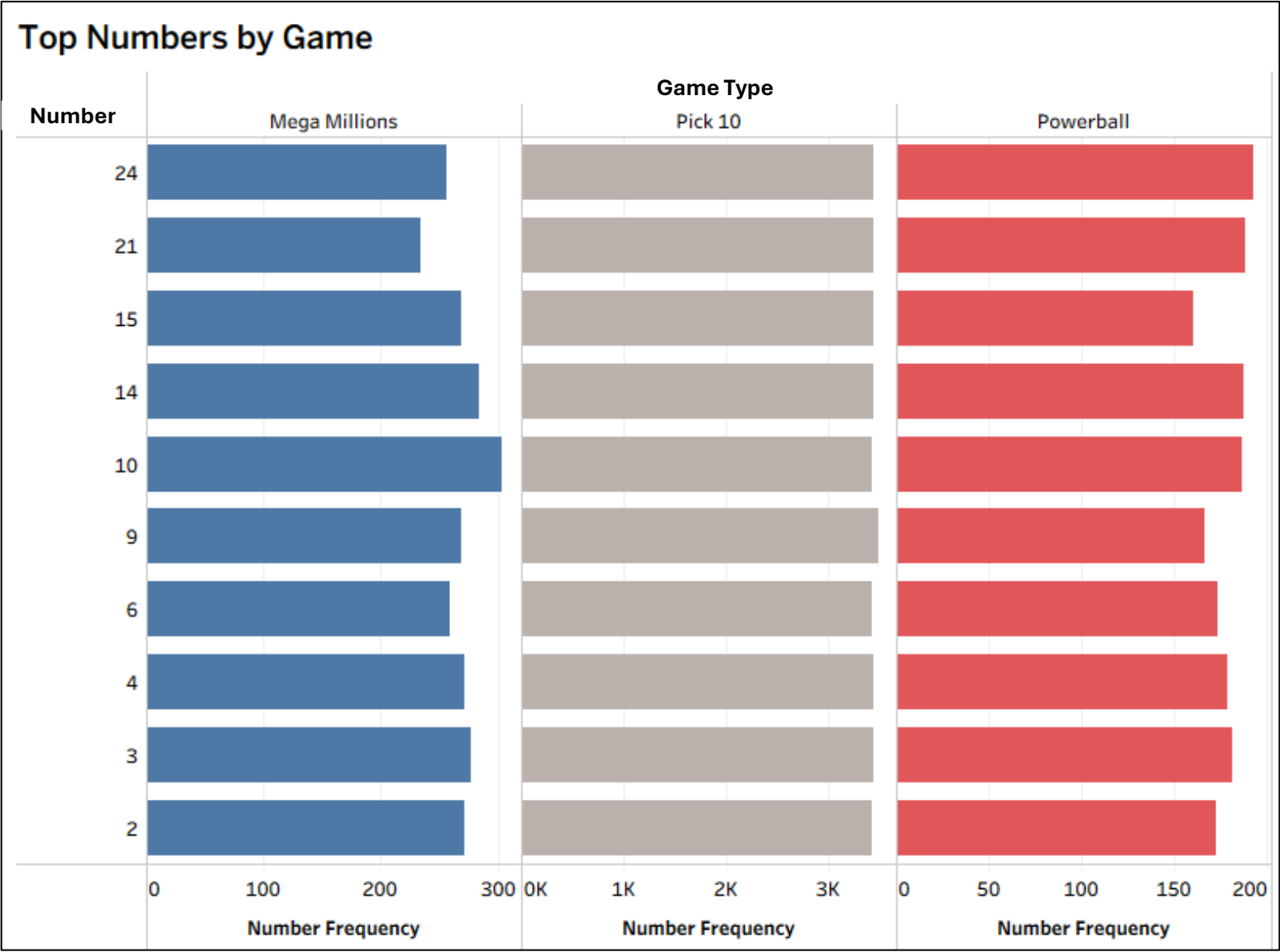
No Clustering in Number Distribution

Top Numbers Show No Consistent Advantage



No number dominates across games

Top Numbers Show No Advantage



Top numbers vary, but none dominate consistently

Key Findings

Lottery numbers do not show stable or predictable patterns

Frequency differences are driven by game structure

No number provides a consistent advantage

What This Means

Do not rely on historical lottery numbers for prediction

Outcomes remain random despite perceived patterns

Decisions should not be based on frequency trends

Ethical Considerations

Data sourced from official dataset (data.ny.gov)

Pick 10 produces more values per draw, impacting comparisons

Partial-year data may affect recent trends

Visual choices could influence interpretation if not clearly explained

No data was altered beyond necessary cleaning and structuring for analysis

All transformations were applied consistently across datasets

Design and Visualization Choices

Simple charts used for clarity (bar, line, histogram)

Consistent color scheme across games

Separation of Pick 10 to avoid misleading comparisons

Clean layout to support readability

Gestalt principles (proximity, similarity, alignment) used to group visuals and improve clarity

Conclusion

While variation exists in lottery number frequency, the data does not support predictable or meaningful patterns

Outcomes remain consistent with randomness

References

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- New York State Open Data. (n.d.). *Lottery Pick 10 Winning Numbers Beginning 1987*. Retrieved from <https://data.ny.gov>
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